

Antiviral Medications for Treatment of Nonsevere Influenza

A Systematic Review and Network Meta-Analysis

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IMPORTANCE The optimal antiviral drug for treatment of nonsevere influenza remains unclear.

OBJECTIVE To compare effects of antiviral drugs for treating nonsevere influenza.

DATA SOURCES MEDLINE, Embase, CENTRAL, CINAHL, Global Health, Epistemonikos, and ClinicalTrials.gov were searched from database inception to September 20, 2023.

STUDY SELECTION Randomized clinical trials comparing direct-acting influenza antiviral drugs to placebo, standard care, or another antiviral drug for treating people with nonsevere influenza.

DATA EXTRACTION AND SYNTHESIS Paired reviewers independently performed data extraction and risk of bias assessment. A frequentist network meta-analysis was performed to summarize the evidence and the certainty of evidence was evaluated using the GRADE approach.

MAIN OUTCOMES AND MEASURES Mortality, admission to hospital, admission to the intensive care unit, duration of hospitalization, time to alleviation of symptoms, emergence of resistance, and adverse events.

RESULTS Overall, 73 trials with 34 332 participants proved eligible. Compared with standard care or placebo, all antiviral drugs had little or no effect on mortality for low-risk patients and high-risk patients (all high certainty). All antiviral drugs (no data for peramivir and amantadine) had little or no effect on hospital admission for low-risk patients (high certainty). For hospital admission in high-risk patients, oseltamivir (risk difference [RD], −0.4%; 95% CI, −1.0 to 0.4; high certainty) had little or no effect and baloxavir may have reduced risk (RD, −1.6%; 95% CI, −2.0 to 0.4; low certainty); all other drugs may have had little or uncertain effect. For time to alleviation of symptoms, baloxavir probably reduced symptom duration (mean difference [MD], −1.02 days; 95% CI, −1.41 to −0.63; moderate certainty); umifenovir may have reduced symptom duration (MD, −1.10 days; 95% CI, −1.57 to −0.63; low certainty); oseltamivir probably had no important effect (MD, −0.75 days; 95% CI, −0.93 to −0.57; moderate certainty). For adverse events related to treatment, baloxavir (RD, −3.2%; 95% CI, −5.2 to −0.6; high certainty) had few or no adverse events; oseltamivir (RD, 2.8%; 95% CI, 1.2 to 4.8; moderate certainty) probably increased adverse events.

CONCLUSIONS AND RELEVANCE This systematic review and meta-analysis found that baloxavir probably reduced risk of hospital admission for high-risk patients and may reduce time to alleviation of symptoms, without increasing adverse events related to treatment in patients with nonsevere influenza. All other antiviral drugs either probably have little or no effect, or uncertain effects on patient-important outcomes.

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+ Supplemental content

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Influenza, a viral respiratory disease caused by influenza virus, affects people worldwide.¹ Although most people with influenza have upper respiratory symptoms that are self-limited and recover within a week without medical attention, some individuals experience severe or fatal complications.^{2,3} This number is substantial: each year, influenza is estimated to cause 3 to 5 million cases of severe illness and 290 000 to 650 000 respiratory deaths worldwide.^{3,4}

Antiviral drugs such as neuraminidase inhibitors (eg, oseltamivir, zanamivir) and selective cap-dependent endonuclease inhibitors (eg, baloxavir) may play a role in reducing illness duration, preventing serious complications, and lowering morbidity and mortality, particularly in high-risk populations.^{5,6} In 2022, the World Health Organization (WHO) published guidelines for clinical management of severe illness caused by influenza virus infections. These guidelines conditionally recommended the use of oseltamivir but advised against using inhaled zanamivir, inhaled laninamivir, or intravenous peramivir for individuals suspected or confirmed to have influenza virus infection with or at risk of severe illness.⁷ Most of the evidence supporting these recommendations was, however, of low or very low certainty, leaving optimal management in doubt.

Previous network meta-analyses have assessed effects of antiviral drugs for treating influenza,⁸⁻¹⁰ but have been limited in failure to provide absolute effects of interventions,⁸⁻¹⁰ in their focus solely on specific categories of interventions such as neuraminidase inhibitors or particular subtypes of influenza such as seasonal influenza,⁸⁻¹⁰ or overlooked crucial patient-important outcomes including mortality and hospitalization.⁸⁻¹⁰ In addition, some studies^{9,10} did not assess the certainty of evidence and none, to our knowledge, have included recent trials.¹¹⁻¹⁴

To evaluate the effectiveness and safety of antiviral treatment for patients with nonsevere influenza and thus support an update of WHO guidelines on antiviral treatment for influenza,⁷ we performed a systematic review and network meta-analysis of randomized clinical trials (RCTs).

Methods

We registered this systematic review protocol with PROSPERO (CRD42023456650) and reported the review according to the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) reporting guidelines for network meta-analyses.¹⁵

Search Strategy and Selection Criteria

With the aid of a medical librarian, we searched MEDLINE, Embase, CENTRAL, CINAHL, Global Health, Epistemonikos, and ClinicalTrials.gov from databases inception to September 20, 2023 (eAppendix 1 in Supplement 1). To identify additional trials, we manually searched reference lists of relevant previous systematic reviews and eligible studies.

We included RCTs that compared direct-acting influenza antiviral drugs to no intervention, placebo, standard care, or another antiviral drug for people with suspected or laboratory-

Key Points

Question What is the optimal antiviral drug in the treatment of patients with nonsevere influenza virus infection?

Findings In this systematic review and network meta-analysis of 73 trials involving 34 332 participants, baloxavir might have reduced the risk of admission to hospital for high-risk patients, probably reduced time to alleviation of symptoms, and did not increase adverse events related to treatment, but might have increased emergence of resistance. Oseltamivir had little or no effect on mortality and admission to hospital, likely had no important effect on time to alleviation of symptoms, and likely increased adverse events related to treatments.

Meaning Baloxavir may be superior to standard care or placebo in reducing the risk of admission to hospital for high-risk patients with nonsevere influenza virus infection and probably decreases time to alleviation of symptoms with few or no adverse effects.

confirmed (by reverse transcription polymerase chain reaction assay, rapid antigen test, or immunofluorescence assay) nonsevere influenza. We used the WHO definitions for disease severity; severe illness from influenza infection was defined as an illness that required hospitalisation.⁷ We defined nonsevere influenza as the absence of any criteria for severe influenza and included RCTs enrolling naturally infected patients with any type of influenza virus (eAppendix 2.1 in Supplement 1). We focused on influenza antiviral drugs approved by the US Food and Drug Administration or other regulatory organizations worldwide, including baloxavir, oseltamivir, laninamivir, zanamivir, peramivir, umifenovir, favipiravir, amantadine, and rimantadine.⁵

Outcomes

Patient-important outcomes of interest included mortality, admission to hospital, admission to intensive care unit (ICU), progression to mechanical ventilation, duration of mechanical ventilation, duration of hospitalization, time to alleviation of symptoms, hospital discharge destination, emergence of resistance, any adverse events, adverse events related to treatments, and serious adverse events. We defined time to alleviation of symptoms as time from the start of treatment to the improvement of all influenza-associated symptoms.^{16,17}

Study Selection, Data Extraction, and Risk of Bias Assessment

Using Covidence, pairs of reviewers (Y.G., Y.Z., M.L., S.L., and X.C.) independently screened titles and abstracts of all records and full texts of potentially eligible records. We checked the retractions for all eligible publications; if the study was retracted from the publication, we excluded the study from our review.¹⁸ Paired reviewers (Y.G., Y.Z., M.L., S.L., Q.Z., J.X., Y.S., W.Z., Z.L., S.H., and J.H.) independently extracted data on study characteristics, patient characteristics, characteristics of interventions, and outcomes (eAppendix 2.2 in Supplement 1) and evaluated the risk of bias of eligible RCTs using a modified Cochrane risk of bias tool (eAppendix 2.3 in Supplement 1).¹⁹ Reviewers resolved discrepancies by discussion or, if necessary, by adjudication with a third reviewer (Q.H.).

Statistical Analysis

We conducted pairwise meta-analyses for all direct comparisons of each outcome using the Hartung-Knapp-Sidik-Jonkman (HKSJ) random-effects model in R (version 4.2.1; R Foundation for Statistical Computing). To assess the between-study heterogeneity, we visually inspected forest plots and used the I^2 statistic. For comparisons that included at least 10 studies, to assess publication bias, we used Harbord test for dichotomous outcomes and Egger test for continuous outcomes,^{20,21} as well as a visual assessment of the funnel plot.

Using netmeta package in R (version 4.2.1; R Foundation), we performed frequentist random-effects network meta-analyses with a graph-theoretical approach.²² The estimator was based on weighted least-square regression using the Moore-Penrose pseudoinverse method.²² For each outcome, we used the design-by-treatment model (global test) to assess the coherence assumption for the entire network.²³ We calculated indirect estimates from the network by node-splitting and a back-calculation method.²⁴ To evaluate local (loop-specific) incoherence between direct and indirect evidence, we used the node-splitting method and calculated a P value for the test of incoherence.²⁵

We used risk ratios (RRs) with 95% CIs for mortality, hospital admission, any adverse events, and adverse events related to treatments. We calculated absolute effects using the network RR estimates and the baseline risk estimates. To estimate absolute effects of antiviral drugs on mortality and hospital admission, the guideline panel recommended use of 2 baseline risk categories for patients at low risk and high risk for severe complications. We defined high-risk patients using the definitions from 2022 WHO influenza guidelines.⁷

For admission to ICU and serious adverse events, the median event rate from the standard care or placebo arm was less than 1%; we directly calculated the absolute risk differences (RDs) with 95% CIs. We calculated mean differences (MDs) with 95% CIs for continuous outcomes. When standard deviations (SDs) were missing, we estimated them using methods described in the Cochrane Handbook.²⁶ eAppendix 2.4 in Supplement 1 provides further details for statistical analysis.

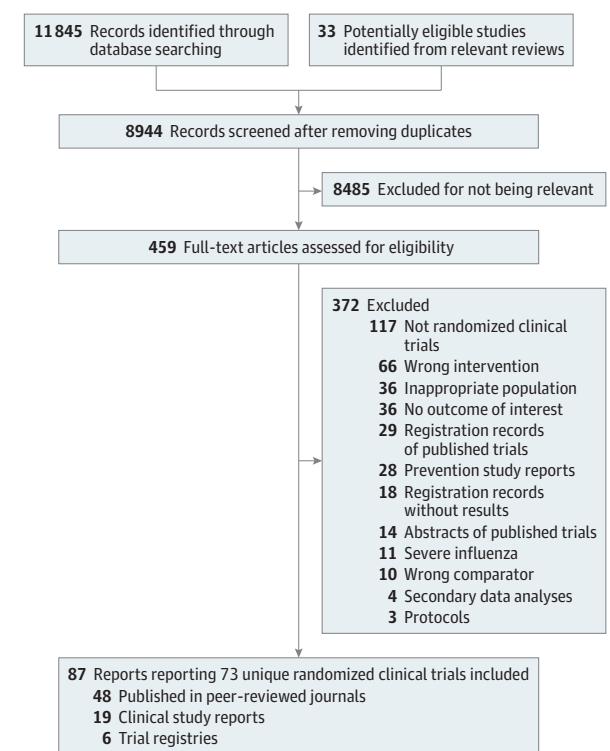
Subgroup Analysis and Sensitivity Analysis

If at least 2 trials provided relevant information for each subgroup, we performed the following prespecified within-trial subgroup analyses: influenza etiology; confirmed vs suspected influenza; age; patients at increased risk of severe complications vs those at low risk (eAppendix 2.5 in Supplement 1). We planned to assess the credibility of the subgroup hypothesis using the Instrument for Assessing the Credibility of Effect Modification Analyses (ICEMAN) tool.²⁷

Certainty of Evidence

To assess the certainty of evidence for each outcome, we used the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach for network meta-analysis.^{28,29} We evaluated the risk of bias, imprecision, inconsistency, indirectness, and publication bias, and rated the certainty of evidence for direct estimates as high, moderate, low, or very low.³⁰ We assessed imprecision at the network level using

Figure 1. Study Selection



the minimal important difference (MID) as a threshold.³¹ We rated imprecision following GRADE guidance (eAppendix 2.6 in Supplement 1).^{32,33} We developed the summary of findings tables for comparisons of each antiviral vs standard care or placebo in MAGICapp following GRADE guidance.^{34,35}

Results

The electronic search identified 11 878 records. After screening 8944 unique titles and abstracts and 459 full texts, 87 articles reporting 73 unique RCTs proved eligible (Figure 1). eAppendix 3 in Supplement 1 lists the eligible trials.

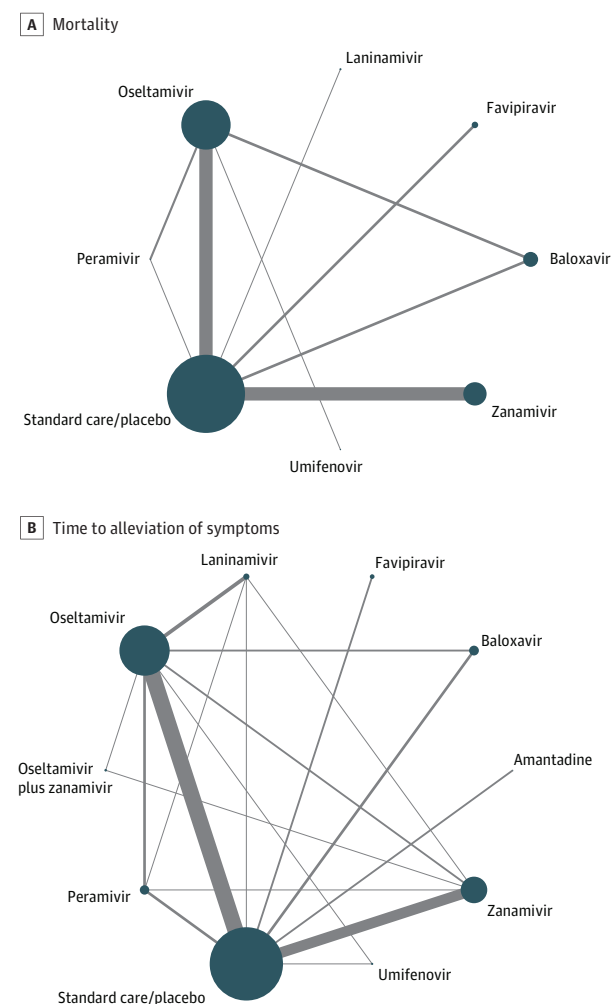
Characteristics of Included Studies

Eligible trials were published during 1971 through 2023 and evaluated 8 unique antiviral drugs: baloxavir, favipiravir, laninamivir, oseltamivir, peramivir, umifenovir, zanamivir, and amantadine, with the most common comparison being between oseltamivir and standard care or placebo. Sixty-five trials included patients within 2 days of symptom onset. Sample sizes ranged from 14 to 3266 (a total of 34 332); the median of mean age was 35.0 years, and the median proportion of men was 49.8% (eAppendix 4 in Supplement 1). The follow-up for outcomes ranged from 5 to 29 days.

Risk of Bias

eAppendix 5 in Supplement 1 presents the risk of bias of eligible trials for each outcome. Nine trials were at low or probably low

Figure 2. Network Plot for Mortality and Time to Alleviation of Symptoms



The size of the circle represents the number of participants. The connecting lines represent direct comparisons. The width of the line represents the number of studies.

risk of bias for all reported outcomes. The main limitations were inadequate allocation concealment and the lack of blinding of data collectors, outcome assessors, and data analysts.

Outcomes

Figure 2 and eAppendix 6 in Supplement 1 present network plots. We did not find substantial between-study heterogeneity for most outcomes and most comparisons (eAppendix 7 in Supplement 1). Tests of incoherence raised no concerns of global incoherence (eAppendix 8 in Supplement 1) for all outcomes and no concerns of local incoherence (eAppendix 9 in Supplement 1) between direct and indirect evidence except for time to alleviation of symptoms. eAppendix 10 in Supplement 1 presents network relative estimates and absolute estimates, with certainty of evidence. Table 1, Table 2, and eAppendix 11 in Supplement 1 present GRADE summary of findings for antivirals vs standard care or placebo. We did not find evidence of intransitivity. We found no evidence of publication bias (eAppendix 12 in Supplement 1).

Mortality

Forty-one trials including 23 892 patients reported mortality. High certainty evidence showed that, compared with standard care or placebo, 7 antiviral drugs (ie, baloxavir, favipiravir, laninamivir, oseltamivir, peramivir, umifenovir, zanamivir) had little or no effect on mortality in low-risk (RD varied from -0.12 to -0.02 per 1000) or high-risk (RD varied from -1.22 to -0.24 per 1000) patients (Table 1, Table 2, Table 3; eAppendix 11 in Supplement 1).

Admission to Hospital

Twenty-eight trials involving 17 262 patients reported admission to hospital for all antivirals except for peramivir and amantadine. For low-risk patients, there proved little or no difference between 6 antiviral drugs (ie, baloxavir, favipiravir, laninamivir, oseltamivir, umifenovir, and zanamivir) and standard care or placebo in hospital admission (RD varied from -2 to 1 per 1000; high certainty). For admission to hospital in high-risk patients, compared with standard care or placebo, baloxavir may have reduced risk (RD, -16 per 1000; 95% CI, -20 to 4 ; low certainty); oseltamivir (RD, -4 per 1000; 95% CI, -10 to 4 ; high certainty) and zanamivir (RD, 4 per 1000; 95% CI, -4 to 15 ; high certainty) had little or no effect; and laninamivir may have had little or no effect (RD, 2 per 1000; 95% CI, -21 to 979 ; low certainty) (Table 1, Table 2, Table 3; eAppendix 11 in Supplement 1).

Admission to ICU

Two trials with 1626 patients provided data on admission to ICU for oseltamivir and peramivir. Compared with standard care or placebo, oseltamivir had little or no effect on admission to ICU (RD, -2 per 1000; 95% CI, -7 to 3 ; high certainty). We are uncertain whether peramivir reduced admission to the ICU (Table 3).

Time to Alleviation of Symptoms

Fifty-nine trials enrolling 24 086 patients reported time to alleviation of symptoms. Compared with standard care or placebo, baloxavir showed an MD in symptom duration of 1.02 days shorter (95% CI, -1.41 to -0.63 ; moderate certainty); umifenovir showed an MD in symptom duration of 1.10 days shorter (95% CI, -1.57 to -0.63 ; low certainty); whereas oseltamivir (MD, -0.75 days; 95% CI, -0.93 to -0.57 ; moderate certainty) and zanamivir (MD, -0.68 days; 95% CI, -0.93 to -0.43 ; moderate certainty) had a mean decrease in symptom duration of less than 1 day (Table 1, Table 2, Table 3; eAppendix 11 in Supplement 1).

Duration of Hospitalization

Three trials with 234 patients provided data on duration of hospitalization for oseltamivir, peramivir, and zanamivir. The evidence was very uncertain as to whether these drugs reduced duration of hospitalization (Table 3).

Adverse Events

Forty-nine trials with 22 868 patients reported data on overall adverse events. Compared with standard care or placebo, baloxavir (RD, -52 per 1000; 95% CI, -87 to -17 ; high certainty) did not increase overall adverse events and zanamivir

Table 1. Summary of Findings for Baloxavir vs Standard Care or Placebo

Outcome	Study results and measurements	Absolute effect estimates per 1000		Absolute difference per 1000 (95% CI)	Certainty of the evidence (quality of evidence)	Summary
		Standard care/placebo	Baloxavir			
Mortality (low-risk patients)	Relative risk: 0.83 (95% CI, 0.14-4.82); based on data from 2144 participants in 2 studies	0.20	0.17	0.03 Fewer (0.17 fewer-0.76 more)	High	Baloxavir has little or no effect on mortality for low-risk patients
Mortality (high-risk patients)	Relative risk: 0.83 (95% CI, 0.14-4.82); based on data from 2144 participants in 2 studies	2	1.66	0.34 Fewer (1.72 fewer-7.64 more)	High	Baloxavir has little or no effect on mortality for high-risk patients
Admission to hospital (low-risk patients)	Relative risk: 0.24 (95% CI, 0.05-1.19); based on data from 1461 participants in 2 studies	3	1	2 Fewer (3 fewer-1 more)	High	Baloxavir has little or no effect on admission to hospital for low-risk patients
Admission to hospital (high-risk patients)	Relative risk: 0.24 (95% CI, 0.05-1.19); based on data from 1461 participants in 2 studies	21	5	16 Fewer (20 fewer-4 more)	Low; due to very serious imprecision	Baloxavir may reduce the risk of admission to hospital for high-risk patients
Adverse events related to treatments	Relative risk: 0.74 (95% CI, 0.57-0.95); based on data from 2776 participants in 3 studies	122	90	32 Fewer (52 fewer-6 fewer)	High	Baloxavir does not increase adverse events related to treatments
Serious adverse events	Risk difference: 0.001 (95% CI, -0.004-0.005); based on data from 2776 participants in 3 studies	5	6	1 More (4 fewer-5 more)	Moderate; due to serious imprecision	Baloxavir likely has little or no effect on serious adverse events
Time to alleviation of symptoms, d	Measured by: day; lower better; based on data from 1855 participants in 3 studies	Mean: 4.92	Mean: 3.90	Mean difference, 1.02 lower (1.41 lower-0.63 lower)	Moderate; due to serious imprecision	Baloxavir likely reduces time to alleviation of symptoms

Table 2. Summary of Findings for Oseltamivir vs Standard Care or Placebo

Outcome	Study results and measurements	Absolute effect estimates per 1000		Absolute difference per 1000 (95% CI)	Certainty of the evidence (quality of evidence)	Summary
		Standard care/placebo	Oseltamivir			
Mortality (low-risk patients)	Relative risk: 0.84 (95% CI, 0.34-2.07); based on data from 12 008 participants in 17 studies	0.20	0.17	0.03 Fewer (0.13 fewer-0.21 more)	High	Oseltamivir has little or no effect on mortality for low-risk patients
Mortality (high-risk patients)	Relative risk: 0.84 (95% CI, 0.34-2.07); based on data from 12 008 participants in 17 studies	2	1.68	0.32 Fewer (1.32 fewer-2.14 more)	High	Oseltamivir has little or no effect on mortality for high-risk patients
Admission to hospital (low-risk patients)	Relative risk: 0.8 (95% CI, 0.54-1.18); based on data from 12 589 participants in 20 studies	3	2	1 Fewer (1 fewer-1 more)	High	Oseltamivir has little or no effect on admission to hospital for low-risk patients
Admission to hospital (high-risk patients)	Relative risk: 0.80 (95% CI, 0.54-1.18); based on data from 12 589 participants in 20 studies	21	17	4 Fewer (95% CI, 10 fewer-4 more)	High	Oseltamivir has little or no effect on admission to hospital for high-risk patients
Adverse events related to treatments	Relative risk: 1.23 (95% CI, 1.10-1.39); based on data from 6782 participants in 12 studies	122	150	28 More (95% CI, 12 more-48 more)	Moderate; due to serious risk of bias	Oseltamivir likely increases adverse events related to treatments
Serious adverse events	Risk difference: 0.0 (95% CI, -0.003-0.002); based on data from 14 718 participants in 22 studies	5	5	0 Fewer (3 fewer-2 more)	Moderate; due to serious risk of bias	Oseltamivir likely has little or no effect on serious adverse events
Time to alleviation of symptoms, d	Measured by: day; lower better; based on data from 9078 participants in 22 studies	Mean: 4.92	Mean: 4.17	Mean difference: 0.75 lower (0.93 lower-0.57 lower)	Moderate; due to serious risk of bias	Oseltamivir likely has no important effect on time to alleviation of symptoms

(RD, -28 per 1000; 95% CI, -48 to -7; low certainty) may not have resulted in an increase (Table 3).

Thirty-six trials with 19 298 patients reported data on adverse events related to treatments. Compared with standard care or placebo, baloxavir (RD, -32 per 1000; 95% CI, -52 to -6; high certainty) did not increase adverse events related to treatment and oseltamivir (RD, 28 per 1000; 95% CI, 12 to 48; moderate certainty) likely increased adverse events related to treatment (Table 3).

Fifty-eight trials involving 32 043 patients provided low to moderate evidence that there may be little or no difference in se-

rious adverse events between baloxavir, favipiravir, laninamivir, oseltamivir, peramivir, or zanamivir and standard care or placebo (Table 1, Table 2, Table 3; eAppendix 11 in Supplement 1).

Emergence of Resistance

Twelve trials enrolling 1554 patients reported the emergence of resistance for participants who received antiviral drugs. Baloxavir may have had an important impact on emergence of drug resistance (percentage, 9.97%; 95% CI, 0.02%-31.79%; low certainty). Zanamivir may have had no important impact on emergence of drug resistance (percentage, 0%; 95% CI, 0%-

Table 3. Summary of Effects of Antivirals vs Standard Care or Placebo

Antivirals	Absolute effect (95% CI)									
	Mortality per 1000		Admission to hospital per 1000		Admission to ICU per 1000	Any adverse events per 1000	Adverse events related to treatments per 1000	Serious adverse events per 1000	Time to alleviation of symptoms, d	Duration of hospitalization, d
	Low-risk patients	High-risk patients	Low-risk patients	High-risk patients						
Baseline risk	0.2	2	3	21	2	346	122	5	4.92	5.52
Baloxavir	-0.03 (-0.17 to 0.76) ^a	-0.34 (-1.72 to 7.64) ^a	-2 (-3 to 1) ^a	-16 (-20 to 4) ^b	No data	-52 (-87 to -17) ^a	-32 (-52 to -6) ^a	1 (-4 to 5) ^d	-1.02 (-1.41 to -0.63) ^d	No data
Favipiravir	-0.09 (-0.19 to 1.63) ^a	-0.86 (-1.92 to 16.30) ^a	-1 (-3 to 24) ^a	-9 (-20 to 171) ^c	No data	-14 (-62 to 45) ^c	-13 (-41 to 24) ^b	-2 (-7 to 4) ^d	-0.46 (-0.96 to 0.03) ^d	No data
Laninamivir	-0.10 (-0.20 to 4.79) ^a	-1.00 (-1.98 to 47.92) ^a	0 (-3 to 163) ^a	2 (-21 to 979) ^b	No data	-38 (-332 to 654) ^c	No data	-4 (-11 to 3) ^d	-0.57 (-1.01 to -0.14) ^b	No data
Oseltamivir	-0.03 (-0.13 to 0.21) ^a	-0.32 (-1.32 to 2.14) ^a	-1 (-1 to 1) ^a	-4 (-10 to 4) ^a	-2 (-7 to 3) ^a	-10 (-28 to 10) ^c	28 (12 to 48) ^d	0 (-3 to 2) ^d	-0.75 (-0.93 to -0.57) ^d	-0.35 (-3.19 to 2.49) ^c
Peramivir	-0.12 (-0.19 to 0.61) ^a	-1.22 (-1.92 to 6.08) ^a	No data	No data	19 (-17 to 55) ^c	-14 (-38 to 17) ^c	-1 (-32 to 39) ^b	0 (-7 to 6) ^b	-0.95 (-1.28 to -0.62) ^b	-1.04 (-3.90 to 1.82) ^c
Umifenovir	-0.03 (-0.20 to 9.26) ^a	-0.30 (-1.96 to 92.62) ^a	-2 (-3 to 17) ^a	-15 (-21 to 120) ^c	No data	-114 (-253 to 235) ^c	-52 (-94 to 45) ^c	No data	-1.10 (-1.57 to -0.63) ^b	No data
Zanamivir	-0.02 (-0.13 to 0.26) ^a	-0.24 (-1.32 to 2.56) ^a	1 (-1 to 2) ^a	4 (-4 to 15) ^a	No data	-28 (-48 to -7) ^b	6 (-12 to 26) ^b	2 (-1 to 6) ^b	-0.68 (-0.93 to -0.43) ^d	-1.10 (-2.30 to 0.10) ^c
Amantadine	No data	No data	No data	No data	No data	No data	No data	No data	-0.78 (-1.30 to -0.26) ^b	No data

Abbreviation: ICU, intensive care unit.

^a High certainty.^b Low certainty.^c Very low certainty.^d Moderate certainty.

11.66%; low certainty). We were uncertain whether oseltamivir and peramivir increased emergence of drug resistance (very low certainty; eAppendix 13 in [Supplement 1](#)).

Subgroup Analysis and Sensitivity Analysis

The within-trial subgroup analyses and meta-regression analyses did not reveal any subgroup effects (eAppendix 14 and eAppendix 15 in [Supplement 1](#)). Our sensitivity analyses using the RD as summary effect measure for mortality and admission to hospital (eAppendix 16 in [Supplement 1](#)), only including studies with patients with confirmed influenza (eAppendix 17 in [Supplement 1](#)), only including studies with patients at high risk (eAppendix 18 in [Supplement 1](#)), and only including studies published in peer reviewed journals (eAppendix 19 in [Supplement 1](#)) showed similar results to primary analyses of all patients.

Discussion

In patients with nonsevere influenza, we found high certainty evidence that compared with standard care or placebo, baloxavir, favipiravir, laninamivir, oseltamivir, peramivir, umifenovir, and zanamivir had little or no effect on mortality for low-risk and high-risk patients, high certainty evidence that oseltamivir and zanamivir had little or no effect on hospital admission for high-risk patients, and low certainty evidence that baloxavir may have reduced the risk of hospital admission for high-risk patients. All antivirals had little or no effect on admission to hospital for low-risk patients. For time to al-

leviation of symptoms, baloxavir likely reduced symptom duration; oseltamivir and zanamivir likely had no important effect; and umifenovir may have reduced symptom duration. Baloxavir did not increase adverse events related to treatment but may have resulted in resistance in approximately 10% of those treated. Oseltamivir likely increased risk of adverse events related to treatment.

This review synthesized all available evidence on benefits and harms of antivirals for patients with nonsevere influenza. The WHO guideline panel provided us with MID values for patient-important outcomes and reviewed the results of our review. We used these MID values to interpret the results, assist our imprecision rating, and draw conclusions.

During the rating of imprecision, we followed GRADE guidance and considered external indirect evidence.^{32,33} For mortality in both low- and high-risk patients and for hospital admission in low-risk patients, where the baseline risk was lower than the MID, the absolute effects between comparisons did not exceed the MID, and no intervention was superior to the other despite the certainty of evidence. Therefore, we rated the certainty of evidence for all comparisons as high certainty.

Several network meta-analyses have evaluated antivirals for treating influenza,^{8-10,36,37} but none assessed all approved antivirals for influenza and none focused on patient-important outcomes of mortality and hospital admission. Compared with previous network meta-analyses,^{8-10,36,37} we included more trials and patients, assessed the certainty of evidence using the GRADE approach, and presented absolute effects. Our findings on baloxavir are consistent with previous network meta-analyses that

baloxavir probably reduces time to alleviation of symptoms without increasing adverse events.^{8,10}

A recent pairwise meta-analysis including 12 studies with 6295 outpatients reported that oseltamivir may not have reduced risk of hospitalization and may have increased nausea and vomiting.³⁸ Our findings on the effects of oseltamivir for nonsevere influenza are consistent with results of this meta-analysis.³⁸ By increasing sample size, our systematic review and meta-analysis has added statistical power and improved the precision of estimates and allowed the conclusion that oseltamivir did not result in an important reduction in symptom duration.

Limitations

The evidence for some antiviral drugs and some outcomes was limited. The panel was also interested in progression to mechanical ventilation, duration of mechanical ventilation, and hospital discharge destination, but no trials reported these outcomes. Because within-trial subgroup information was limited, we were only able to conduct within-trial subgroup analysis for certain outcomes and comparisons and did not find any significant subgroup effects.

For some patient-important outcomes (eg, mortality, hospitalization), event rates are often very low. Many included studies may lack the power to detect these outcomes, and even pooled analyses may have insufficient statistical power to reliably evaluate antiviral effects.

There is little empirical evidence supporting the WHO guideline development group's estimates of MIDs for the outcomes of interest. Thus, one might disagree with their estimates and these disagreements could result in importantly different inferences. For instance, if one believes that patients place a high value on a reduction of duration of symptoms of 0.5 days, one would include that oseltamivir results in an important reduction in symptom duration.

We evaluated the emergence of antiviral resistance with the help of a clinical expert, despite it being a lower-ranked outcome. However, antiviral resistance is complex to interpret due

to its rare reporting and the varying clinical significance of different variants.

There was considerable heterogeneity among the eligible studies, including differences in patient populations, time to treatment initiation from symptom onset, drug doses, and treatment duration. Clinicians and patients should consider these sources of heterogeneity when interpreting our findings and their applicability to different clinical scenarios. Due to the high risk of bias or serious imprecision, the certainty of evidence was low for many comparisons.

Our findings support the use of baloxavir for treatment of high-risk patients with nonsevere influenza. However, drug resistance monitoring may be needed because baloxavir may increase emergence of drug resistance. Our studies identify evidence gaps regarding antiviral drugs for nonsevere influenza. Future studies could focus on patient-important outcomes (eg, admission to ICU, duration of hospitalization) and test important subgroup analyses such as the influenza virus type, age, and time from onset of symptoms to treatment.

Conclusions

In patients with nonsevere influenza, compared with standard care or placebo, the RD in high-risk patients being admitted to hospital was greatest with baloxavir (−16 per 1000) compared with −4 per 1000 patients for oseltamivir and 4 per 1000 patients for zanamivir, but the wide CI suggests low certainty of this effect. Baloxavir likely reduces time to alleviation of symptoms by 1.02 days, does not increase adverse events related to treatment, but it may be associated with the emergence of resistance in approximately 10% of those treated. Oseltamivir was found to have little or no effect on mortality and admission to hospital, likely has no important effect on time to alleviation of symptoms, and likely increases risk of adverse events related to treatment. All other antiviral drugs are either likely not to be beneficial or have uncertain effects on patient-important outcomes.

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Editor's Note

The Limited Role for Antiviral Therapy in Influenza

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Influenza is associated with substantial morbidity and mortality. Given this, antiviral therapy is commonly prescribed in an effort to prevent these outcomes. We therefore found it surprising that antivirals seem to make little difference for individuals with influenza in outpatient settings, even in the optimized setting of randomized clinical trials.¹

Guidance from the US Centers for Disease Control and Prevention and others generally encourage treatment of influenza with antivirals.² For outpatients with risk factors, as well as their household contacts, antiviral therapy is recommended on the basis that it might help. For outpatients without risk factors, antiviral therapy may nonetheless be prescribed based on clinical judgment. In either case, pressure to start treatment as early as possible means that antivirals are often prescribed without diagnostic testing for influenza or thoughtful consideration of their benefits and risks.

Gao et al¹ provide evidence that questions the current approach. In a large systematic review and meta-analysis of multiple clinical trials of outpatients with influenza, they found that oseltamivir causes more adverse effects (primarily nau-

sea and vomiting) but has little to no effect on relevant clinical outcomes compared with placebo. Baloxavir marboxil, which is used less frequently, has minimal adverse effects and may reduce the duration of symptoms by 1 day, but increases the risk of antiviral resistance. Thus, baloxavir marboxil may be preferable when treatment is needed, but use may diminish effectiveness over time.

Both oseltamivir and baloxavir marboxil may have significant out-of-pocket costs for patients. Oseltamivir is covered by most insurance plans but may require a substantial copay. Baloxavir marboxil is less frequently covered by insurance and is typically more expensive than oseltamivir. No generic form of baloxavir marboxil is available.

When considering the evidence, we are left to wonder why reflexive treatment of influenza with oseltamivir has become so ingrained in clinical practice. One explanation is initial publication bias arising from nonpublication of early negative trials.³ We also think that action bias, or the tendency to do something even when the potential benefits are unclear, could be driving unnecessary use. Regardless, the accumulating evidence indicates that the time has come for clinicians and health care systems to reconsider their basic approach to outpatients with influenza.

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